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3.2.2 How the Macroeconomy Works

The Circular Flow of Income

The circular flow of income is a model showing how money flows continuously between households and firms in the economy. Households supply factors of production (labour, land, capital, enterprise) to firms and receive factor incomes (wages, rent, interest, profit). Firms use these factors to produce goods and services, which households buy with their income.

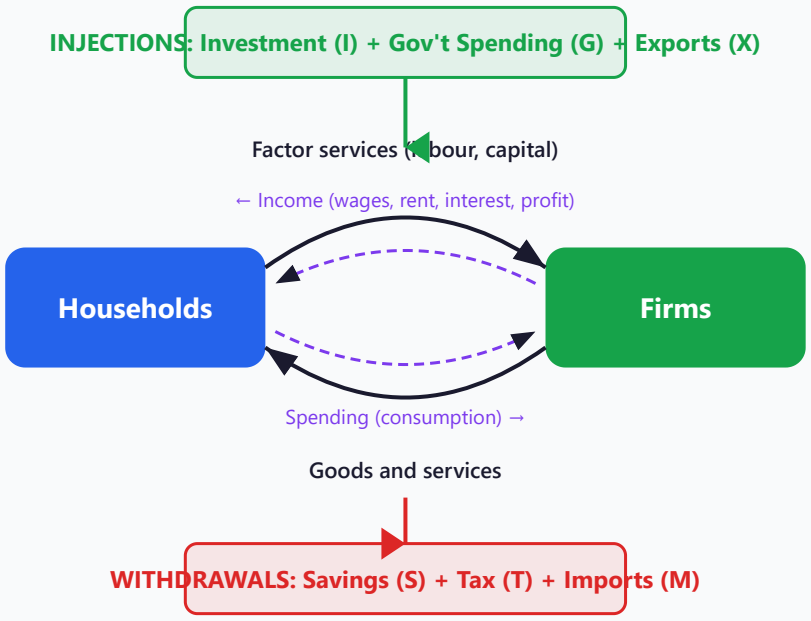

Figure 1: The circular flow of income with injections (J) and withdrawals (W)

Injections and Withdrawals (Leakages)

Injections (J) — money entering the flow	Withdrawals (W) — money leaving the flow

Investment (I): Spending by firms on capital goods (machinery, technology, buildings)	**Savings (S):** Income not spent on consumption (held in banks, pensions, etc.)
Government spending (G): Public spending on services, infrastructure, wages of public sector workers	**Taxation (T):** Income tax, VAT, corporation tax — removed from the flow by government
Exports (X): Spending by foreigners on domestically produced goods/services — money flows in from abroad	**Imports (M):** Spending by domestic consumers on foreign goods/services — money flows out abroad

Equilibrium in the Circular Flow

$$\text{National income is in equilibrium when: } J = W \text{ (i.e., } I + G + X = S + T + M\text{)}$$

- If **J > W**: more money is entering than leaving → national income **rises** → the economy grows
- If **W > J**: more money is leaving than entering → national income **falls** → the economy contracts
- If **J = W**: national income is in equilibrium (stable, not necessarily at full employment)

Aggregate Demand (AD)

Aggregate Demand (AD) is the total planned expenditure on goods and services produced in an economy at a given price level, over a given time period. It is the demand side of the macroeconomy.

$$AD = C + I + G + (X - M)$$

Component	What It Is	Approx. Share of UK AD
C — Consumption	Household spending on goods and services (food, rent, entertainment, transport)	~60% (the largest component)

I — Investment	Spending by firms on capital goods (machinery, factories, technology, vehicles). Also includes changes in inventories.	~16-17%
G — Government spending	Government expenditure on public services and infrastructure (NHS, education, defence, roads). Excludes transfer payments (benefits/pensions).	~20%
(X − M) — Net exports	Exports (foreign spending on UK goods/services) minus imports (UK spending on foreign goods/services)	Typically negative (the UK is a net importer)

The AD Curve

The AD curve shows the relationship between the **general price level** and the **total quantity of real output demanded**. It slopes downward from left to right.

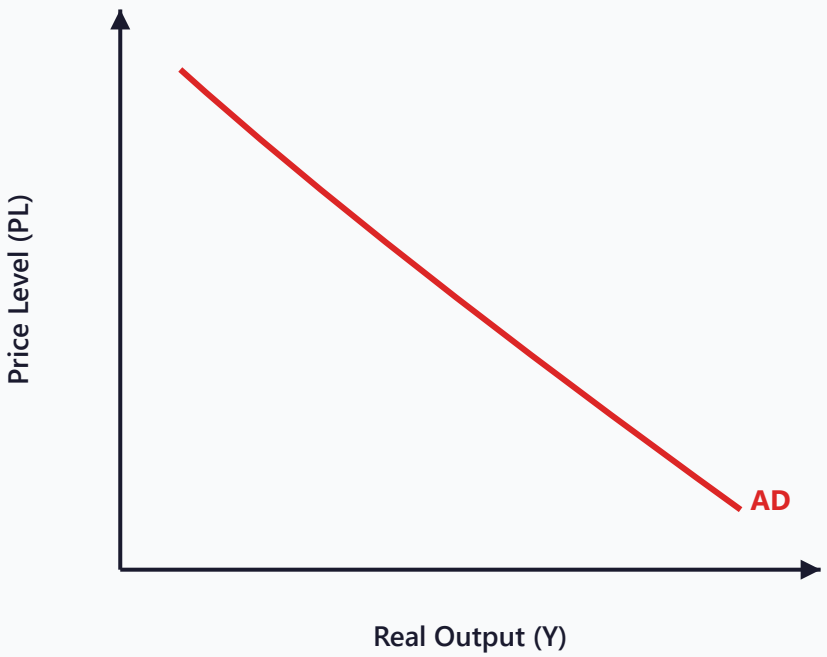

Figure 2: The aggregate demand curve slopes downward — a lower price level is associated with higher real output demanded

Why AD Slopes Downward (Three Effects)

1. **Real balance (wealth) effect:** A higher price level reduces the real value of people's savings and wealth. They feel poorer and spend less → lower C → lower AD.
2. **Interest rate effect:** A higher price level increases the demand for money (people need more money for transactions). This pushes up interest rates, which discourages borrowing and spending → lower C and I → lower AD.
3. **International competitiveness effect:** A higher domestic price level makes UK exports more expensive for foreigners and makes imports cheaper for UK consumers. So exports fall and imports rise → net exports (X−M) falls → lower AD.

Shifts of the AD Curve

AD shifts **right** (increase in AD) when any component increases at every price level. AD shifts **left** when any component decreases.

Component	Factors Shifting AD Right (increase)	Factors Shifting AD Left (decrease)
C (Consumption)	Lower interest rates, rising consumer confidence, rising house/asset prices (wealth effect), tax cuts (more disposable income), easier credit availability	Higher interest rates, falling confidence, falling house prices, tax increases, credit restrictions
I (Investment)	Lower interest rates, improved business confidence ("animal spirits"), technological advances, corporate tax cuts, accelerator effect (rising demand)	Higher interest rates, pessimism, political uncertainty, higher taxes on profit
G (Government spending)	Expansionary fiscal policy (e.g., increased NHS spending, infrastructure projects)	Austerity / contractionary fiscal policy (spending cuts)
(X−M) Net exports	Depreciation of the exchange rate, growth in trading partners' economies, improved competitiveness	Appreciation of the exchange rate, recession in trading partners, loss of competitiveness

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Real-World Example: UK AD Shifts in 2020-21

During COVID-19, AD shifted sharply **left** in 2020: consumer spending collapsed (shops, restaurants, travel all shut), investment plummeted (extreme uncertainty), and exports fell. The government responded with massive fiscal stimulus — the furlough scheme (£70bn), business grants, and eat-out-to-help-out — all designed to shift AD back to the right. In 2021, as restrictions eased, pent-up consumer demand combined with continued government spending caused AD to shift sharply **right**, contributing to the inflation surge of 2022.

The Consumption Function and Key Concepts

The Keynesian consumption function shows the relationship between disposable income (Y) and consumer spending (C):

$$C = a + bY$$

- **a** = autonomous consumption — spending that occurs regardless of income level (funded by savings or borrowing). Even with zero income, people still need to eat and pay rent.
- **b** = marginal propensity to consume (MPC) — the proportion of each additional £1 of income that is spent
- **Y** = disposable income (income after taxes and benefits)

Marginal Propensity to Consume (MPC) and Save (MPS)

Marginal Propensity to Consume (MPC) = ΔC ÷ ΔY — the fraction of each extra pound of income that is spent. E.g., if income rises by £100 and spending rises by £80, MPC = 0.8.

Marginal Propensity to Save (MPS) = ΔS ÷ ΔY — the fraction saved. MPC + MPS = 1 (in a simple model).

In reality, there are also marginal propensities to tax (MPT) and import (MPM). Together, these form the **marginal propensity to withdraw (MPW)**:

$$\text{MPW} = \text{MPS} + \text{MPT} + \text{MPM}$$

$$\text{MPC} + \text{MPW} = 1$$

Factors Influencing Consumption (What Drives the Largest Component of AD)

Factor	Effect on Consumption	Explanation
Disposable income	↑ income → ↑ C	The most important determinant (Keynes). Higher income means more money available to spend.
Interest rates	↑ rates → ↓ C	Higher rates increase the cost of borrowing and the reward for saving, discouraging spending. Mortgage holders have less disposable income.
Consumer confidence	↑ confidence → ↑ C	If consumers are optimistic about the future (job security, income growth), they spend more freely.
Wealth effects	↑ asset prices → ↑ C	Rising house prices and share prices make consumers feel wealthier, encouraging them to spend more (and save less).
Credit availability	Easier credit → ↑ C	If banks are willing to lend freely (low deposits, easy approvals), consumption increases — especially for big-ticket items (cars, furniture).
Income distribution	More equal → ↑ C	Lower-income households have a higher MPC. Redistributing income towards them increases total consumption.
Taxation	↑ tax → ↓ C	Higher income tax reduces disposable income directly. Higher VAT raises prices, reducing real spending power.

Factors Influencing Investment

Factor	Effect on Investment	Explanation
Interest rates	↑ rates → ↓ I	Higher rates increase the cost of borrowing to fund investment projects. Projects that were profitable at 3% interest may not be at 6%.
Business confidence	↑ confidence → ↑ I	Keynes called this "animal spirits." If firms are optimistic about future demand, they invest. If pessimistic, they hold back — even if interest rates are low.
Economic growth rate	↑ growth → ↑ I	The **accelerator effect**: rising consumer demand requires firms to expand capacity. A growing economy incentivises investment in new factories and equipment.
Retained profits	↑ profits → ↑ I	Firms often fund investment from retained profits rather than borrowing. Higher profits provide more internal funding.
Government policy	Incentives → ↑ I	Investment tax allowances, grants, and subsidies (e.g., R&D tax credits) can boost business investment.
Technological change	New tech → ↑ I	Firms must invest to adopt new technology (AI, automation) to stay competitive.
Spare capacity	Spare capacity → ↓ I	If firms already have unused capacity, there's less need to invest in more.

Real-World Example: UK Business Investment After Brexit

UK business investment stagnated after the 2016 Brexit referendum. Uncertainty about future trading relationships, regulations, and market access made firms reluctant to commit to long-term investments. Between 2016 and 2023, UK business investment grew by only about 5% in total — compared to roughly 20-25% growth in other G7 countries. This illustrates how **confidence and uncertainty** can be more powerful drivers of investment than interest rates alone.

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Aggregate Supply (AS)

Aggregate Supply (AS) is the total output of goods and services that all firms in an economy are willing and able to produce at a given price level, in a given time period.

Short-Run Aggregate Supply (SRAS)

The SRAS curve slopes **upward** — as the price level rises, firms produce more output. This is because in the short run, many input costs (especially wages) are "sticky" — they don't adjust immediately. So when prices rise, profit margins increase, incentivising firms to expand production.

What Shifts SRAS?

SRAS shifts when the **costs of production** change (at every price level):

SRAS Shifts Left (costs rise)	SRAS Shifts Right (costs fall)
Rising wages	Falling wages or increased productivity
Rising raw material / commodity prices (oil, gas, metals)	Falling commodity prices
Rising energy costs	Cheaper energy
Exchange rate depreciation (imported inputs cost more)	Exchange rate appreciation (imports cheaper)
Higher indirect taxes (VAT increase)	Lower indirect taxes or new subsidies
Supply shocks (pandemic, war, natural disaster)	Favourable supply conditions

Real-World Example: The 2022 Energy Crisis

Russia's invasion of Ukraine in February 2022 caused natural gas prices to spike by over 400% in Europe. Gas is a critical input for electricity generation, manufacturing, heating, and fertiliser production. This was a massive negative supply shock — SRAS shifted sharply left across Europe.

> The result was "stagflation": rising prices (cost-push inflation reaching 10%+ in the UK) combined with stagnating or falling output. This is the textbook SRAS shift left scenario.

Long-Run Aggregate Supply (LRAS)

LRAS represents the economy's **productive potential** — the maximum sustainable output when all resources are fully and efficiently employed. There are two competing views:

The Classical View

LRAS is **vertical** at the full-employment level of output (Yfe). In the long run, the economy always returns to full employment through flexible wages and prices. Changes in AD only affect the price level — not real output. The economy is self-correcting.

The Keynesian View

LRAS has **three sections**:

1. **Horizontal (Keynesian) section:** When the economy has significant spare capacity, output can increase with NO increase in the price level (unemployed resources put to use)
2. **Upward-sloping section:** As the economy approaches full capacity, bottlenecks emerge — some industries hit capacity limits, skilled workers become scarce. Output can still increase, but with rising prices.
3. **Vertical (Classical) section:** At full capacity, the economy cannot produce more regardless of price. Any increase in AD only causes inflation.

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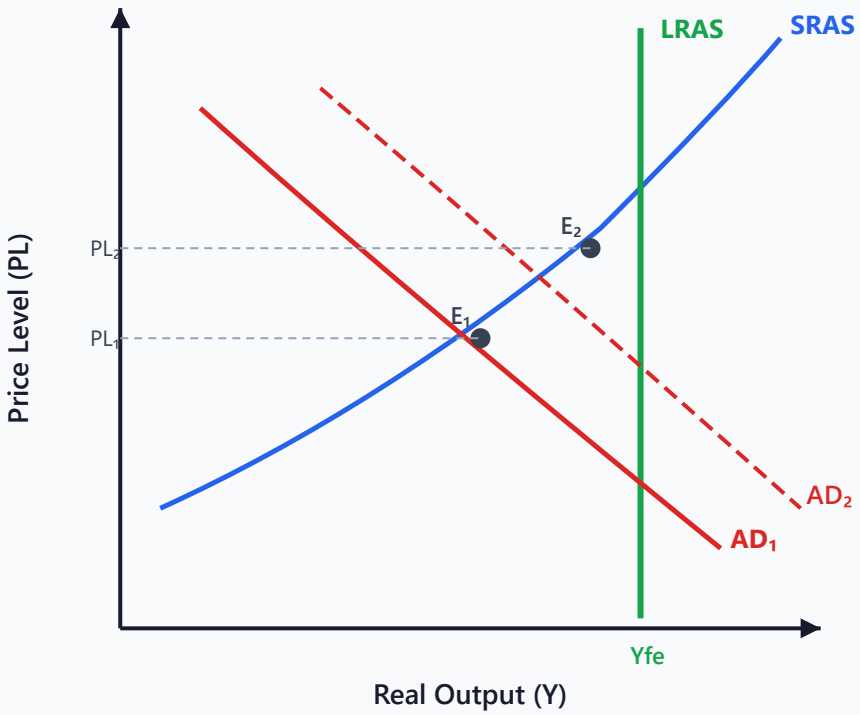

Figure 3: The AD/AS model with Classical (vertical) LRAS. An increase in AD from AD_1 to AD_2 causes both output and the price level to rise in the short run.

What Shifts LRAS? (Increasing Productive Capacity)

LRAS shifts right when the economy's **potential output** increases. This represents genuine long-run economic growth:

Factor	How It Shifts LRAS Right	Example
Quantity/quality of labour	More workers or better-skilled workers increase output potential	Immigration, improved education, vocational training, higher retirement age
Quantity/quality of capital	More or better machinery, technology, and infrastructure	Business investment in automation, government infrastructure spending (roads, broadband)

Technological progress	New methods allow more output from the same inputs	AI, robotics, renewable energy technology, biotech
Natural resources	Discovery or access to new resources	North Sea oil discovery (1970s), fracking revolution in the US
Institutional improvements	Better governance, rule of law, property rights, and market efficiency	Deregulation, reduced corruption, stronger contract enforcement
Enterprise and competition	More entrepreneurs and competitive markets drive innovation and efficiency	Startup culture, removal of barriers to entry

The Multiplier Effect

The multiplier effect describes how an initial injection of spending into the circular flow leads to a **larger final increase** in national income. This happens because one person's spending becomes another person's income, which is then partly spent again, creating successive rounds of spending and income generation.

$$\text{Multiplier (k)} = 1 \div (1 - \text{MPC}) = 1 \div \text{MPW}$$

How the Multiplier Works — Step by Step

Imagine the government invests £100 million in building a new hospital, and the MPC is 0.8 (80% of extra income is spent):

Round	Spending	Saved/Withdrawn	What Happens
Initial injection	£100m	—	Construction workers and suppliers receive £100m of income

Round 1	£80m	£20m	Workers spend 80% of their new income in shops, restaurants, etc.
Round 2	£64m	£16m	Shop owners and restaurant workers spend 80% of their new income
Round 3	£51.2m	£12.8m	And so on, with each round getting smaller...
Total increase in national income			**£500m** (= £100m × multiplier of 5)

$$\text{Multiplier} = 1 \div (1 - 0.8) = 1 \div 0.2 = 5$$

Factors Affecting the Size of the Multiplier

Factor	Effect on Multiplier Size
Higher MPC (people spend more of extra income)	Larger multiplier
Higher marginal rate of tax (more withdrawn at each stage)	Smaller multiplier
Higher marginal propensity to import (spending leaks abroad)	Smaller multiplier
Higher marginal propensity to save	Smaller multiplier

In practice, the UK multiplier is estimated at around **1.0 to 1.5** for government spending — much lower than the theoretical example above. This is because the UK has high tax rates, high import propensity (open economy), and financial markets may offset government spending (crowding out).

The Negative (Reverse) Multiplier

The multiplier works in **both directions**. A withdrawal from the circular flow (e.g., government spending cuts, a fall in investment) causes a multiplied fall in national income. This is why recessions can be self-reinforcing: job losses → less spending → more job losses → even less spending.

Real-World Example: UK Austerity (2010-2015)

After 2010, the UK government pursued austerity — significant cuts to public spending to reduce the budget deficit. Critics argued (using multiplier theory) that cutting spending during a weak recovery would cause a multiplied fall in national income, slowing growth further. Supporters argued the multiplier was small and that reducing the deficit would boost confidence. UK growth was indeed sluggish during 2010-12 (around 1-2%), though it accelerated later. The debate about the size of the fiscal multiplier during recessions remains one of the most contested issues in macroeconomics.

Evaluation of the Multiplier:

- **Time lags:** The multiplier process takes time — the full effect may take 12-18 months to materialise
- **Crowding out:** If government spending is financed by borrowing, it may push up interest rates, reducing private investment — partially or fully offsetting the multiplier
- **Inflation risk:** If the economy is near full capacity, the multiplied increase in AD may cause inflation rather than real growth
- **Leakages:** In an open economy like the UK, much of the extra spending leaks out as imports, reducing the multiplier's domestic impact
- **Confidence effects:** The multiplier assumes a predictable MPC, but consumer and business confidence can change unpredictably
- **Despite these limitations:** The multiplier remains central to understanding why recessions are self-reinforcing and why fiscal stimulus can be effective (particularly when the economy has significant spare capacity)

AD/AS Analysis: Key Macroeconomic Scenarios

1. Demand-Pull Inflation

Cause: AD shifts right when the economy is already near or at full capacity.

Result: The price level rises significantly, but real output increases only slightly (or not at all if at Yfe). "Too much money chasing too few goods."

Examples: Consumer credit boom, large tax cuts when unemployment is already low, quantitative easing causing excess liquidity.

2. Cost-Push Inflation

Cause: SRAS shifts left due to rising production costs (supply shock).

Result: The price level rises AND real output falls simultaneously — this combination is called **stagflation** (stagnation + inflation). It is the worst of both worlds.

Examples: 1970s oil crisis, 2022 energy price shock, pandemic supply chain disruptions.

3. Long-Run Economic Growth

Cause: LRAS shifts right (increased productive capacity through investment, technology, education).

Result: The economy can produce more output at every price level. If AD also increases proportionally, the economy grows with stable or falling prices — the ideal outcome.

Examples: China 1990-2020, post-WWII Japan, technology-driven growth in the US in the 1990s.

4. Recession / Negative Demand Shock

Cause: AD shifts left (falling consumer confidence, investment collapse, financial crisis).

Result: Real output falls, unemployment rises, the price level may fall (or inflation slows). In the Keynesian view, the economy can get stuck in this position without government intervention — there is no automatic self-correction.

Examples: 2008 financial crisis, 2020 COVID lockdowns.

> **Exam Tip:** For any macro question, draw the AD/AS diagram. Show the initial equilibrium, the shift, and the new equilibrium. Label everything clearly: initial and new price levels (PL_1, PL_2), output levels (Y_1, Y_2), and the shift arrow. Then explain the adjustment process in words. This structured approach scores consistently high marks.